MMET 410 – Lab 8

Instructions

For task 1/2/3/4/5, check with TA before you close any circuit, and include answers in your lab report.

For task 6, include four parts as usual- description, action sequence, IN/OUT address, and program screenshot

Due date for a lab report is tomorrow midnight

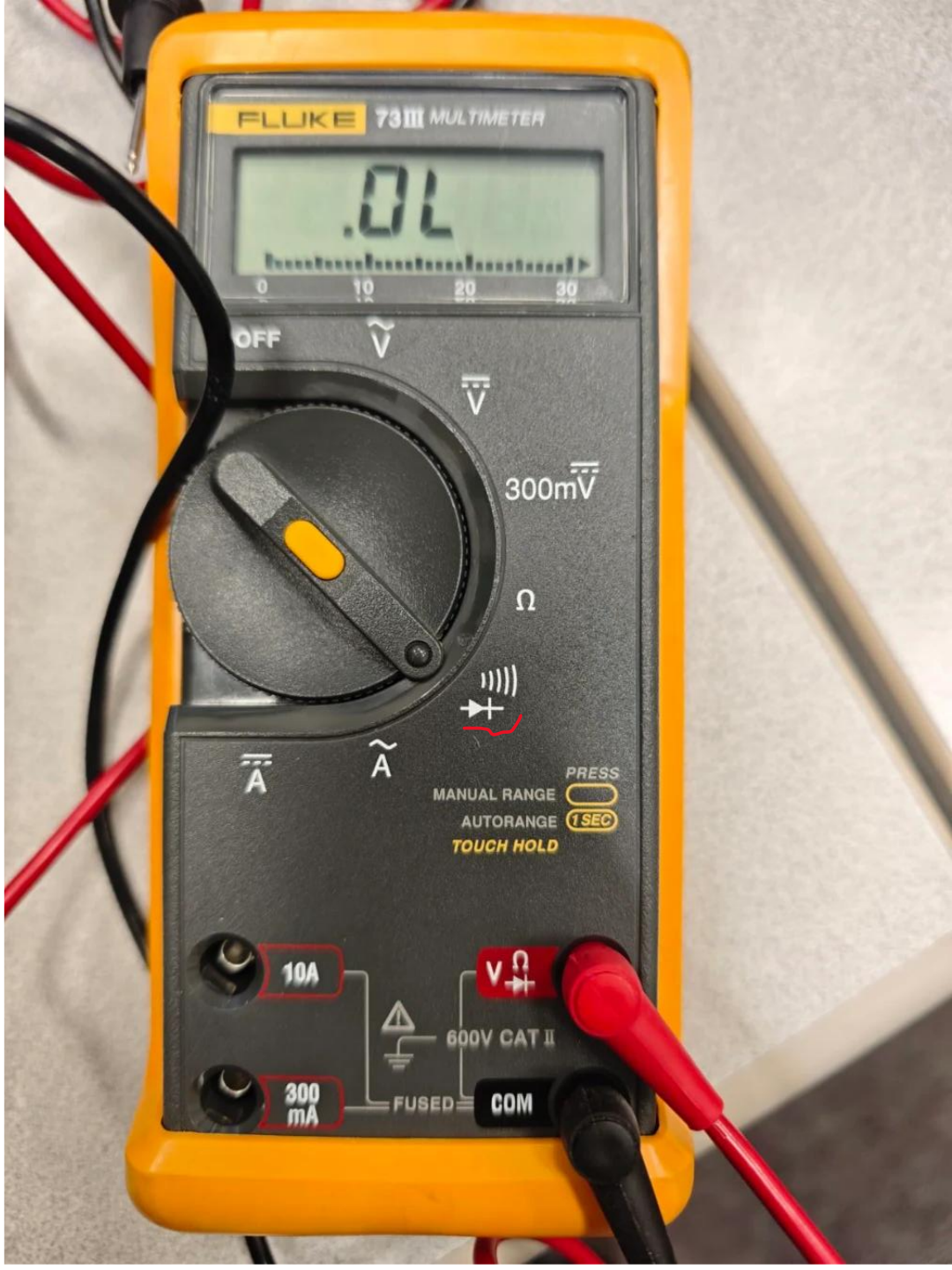
Task 1 - 5

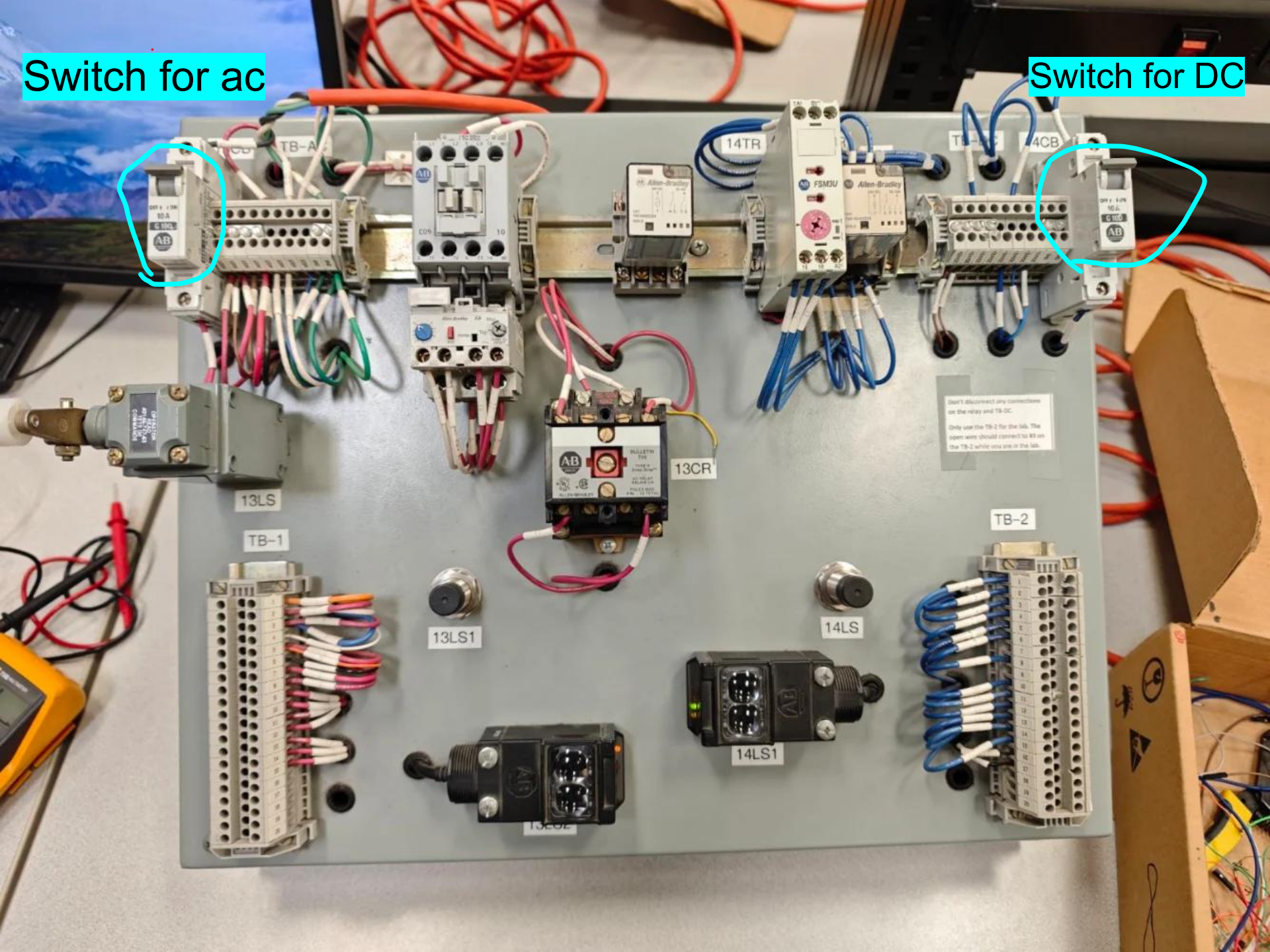
- Work on the connections as per the lab manual
- Answer the questions given.

Don’t connect the motor to the supply! It only works with 5V

Use a 5V provider

Use multimeter to test close or open




Switch for ac
Switch for DC
TB-A
14TR
4CB
13LS
13CR
TB-1
TB-2
13LS1
14LS
14LS1

Do not disconnect any wires.

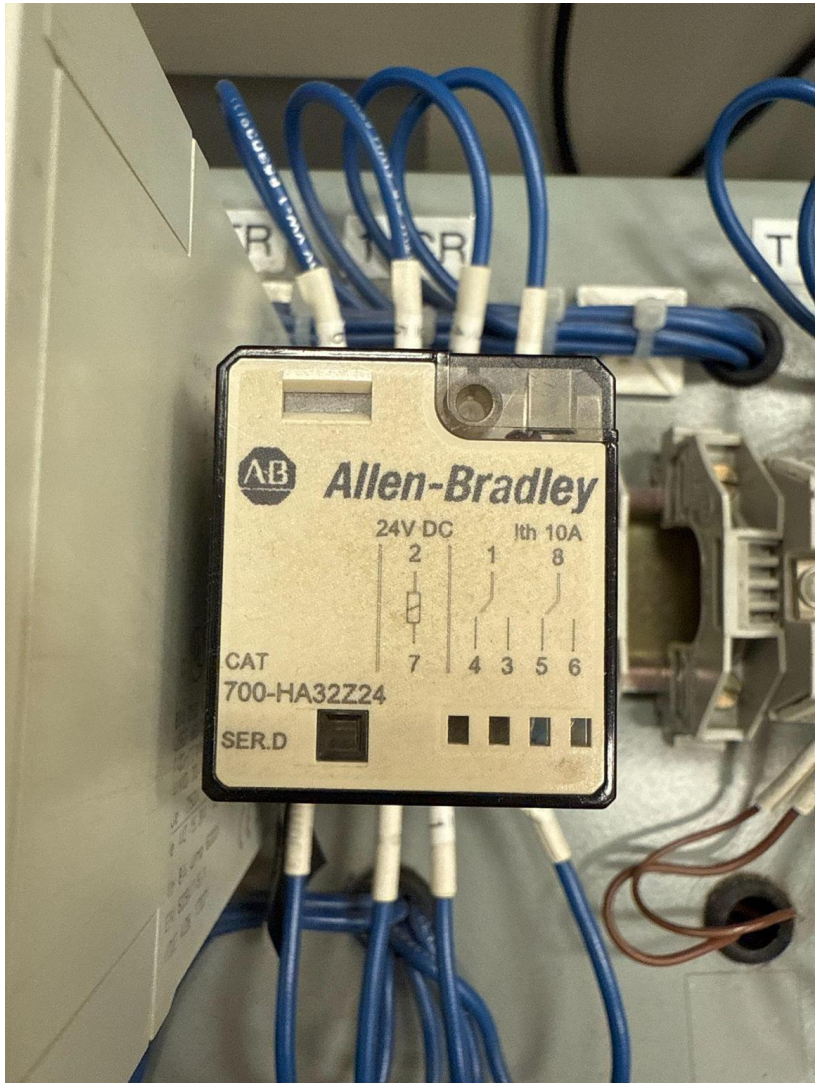
Ask me if you want to turn switch on.

Ask me to turn on the switch.

Relay

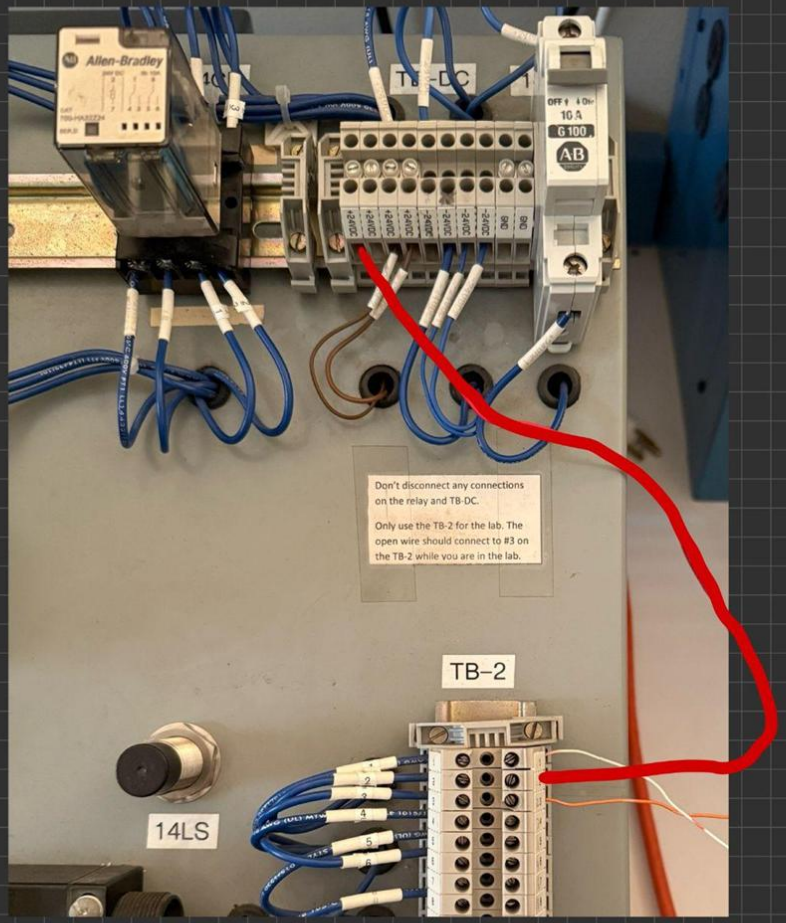
Relay has two circuits, it uses one circuit to control the contact of another circuit.

- Energizing the coil
- Generate a magnetic field
- Attract or repel the magnetic material inside the relay
- Open or close the circuit



Task 1b.

Find an empty 24V



3: Interface ControlLogix 5550 output terminals, relay coil, and power supply (+-24VDC)

a. Connect +24VDC to terminal #6 of slot 7; connect the wire from terminal #5 of slot 7 to terminal #2 of the coil in the relay

X b. Connect -24VDC to terminal #7 of the coil in the relay.

Have your instructor check the wiring before you turn on the power supply switch!

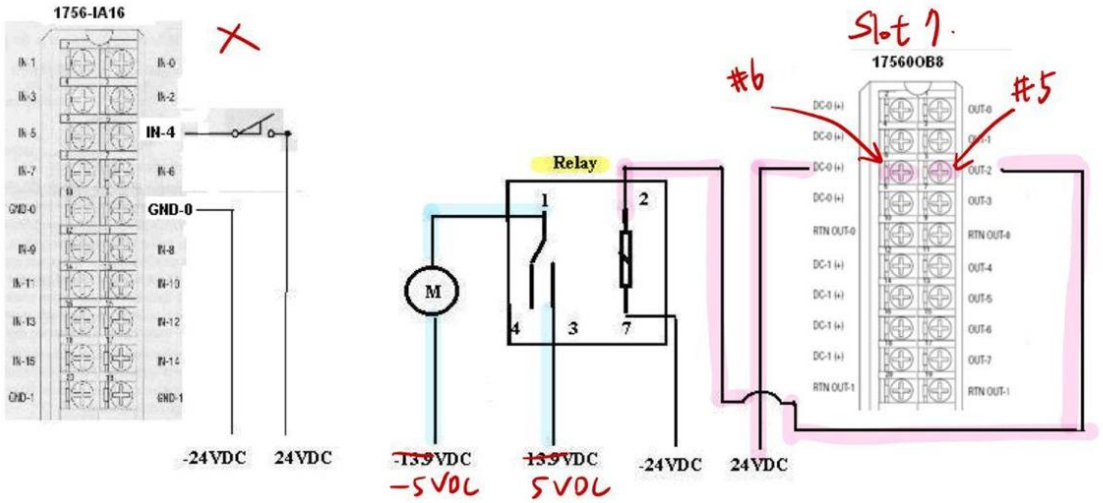
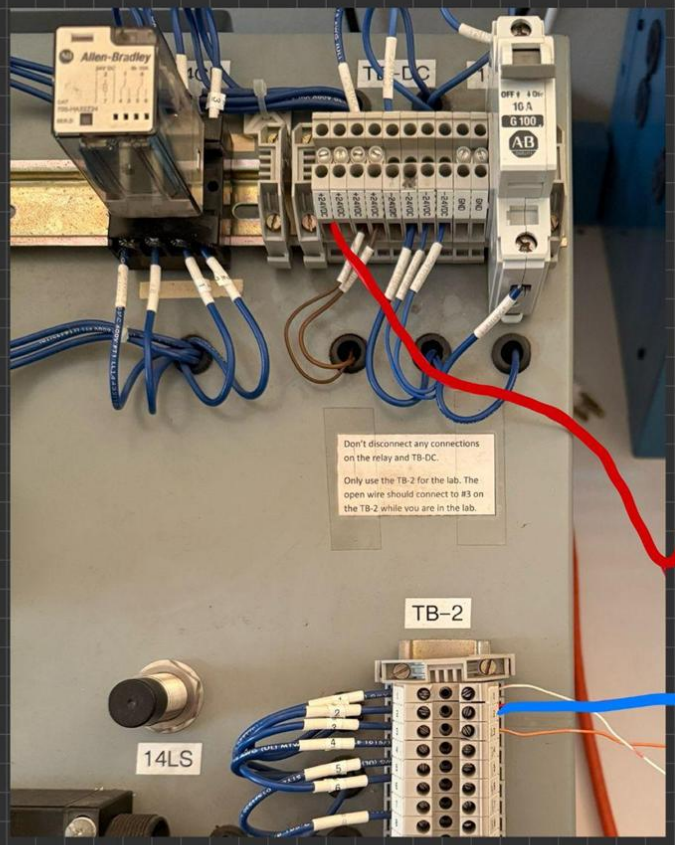
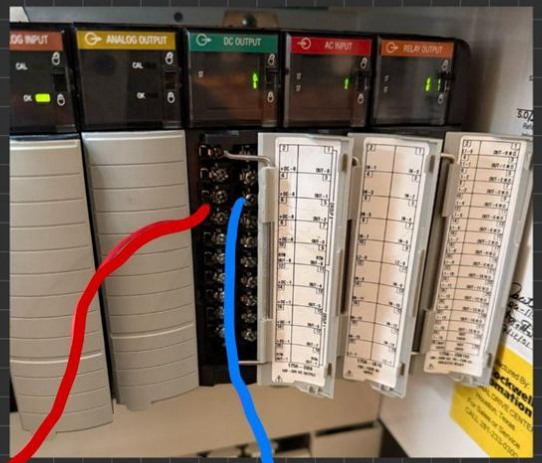


Figure1 shows how the relay, motor, and 1756OB8 module been interfaced.

Task 3a.


Allen-Bradley
Don't disconnect any connections on the relay and TB-DC.
Only use the TB-2 for the lab. The open wire should connect to #3 on the TB-2 while you are in the lab.
TB–2
14LS


ANALOG OUTPUT
DC OUTPUT
AC INPUT
RELAY OUTPUT

4: Test the functionality of the motor. 5V

a. Using the given power supply (~~13.9~~ V), connect the wires to the motor:
   Is the motor running?_____.
b. Following Figure 1 schematic below to interface the motor with normally open (N.O.) terminals of the relay (i.e.terminal #3) and power supply. Is the motor running? _____.

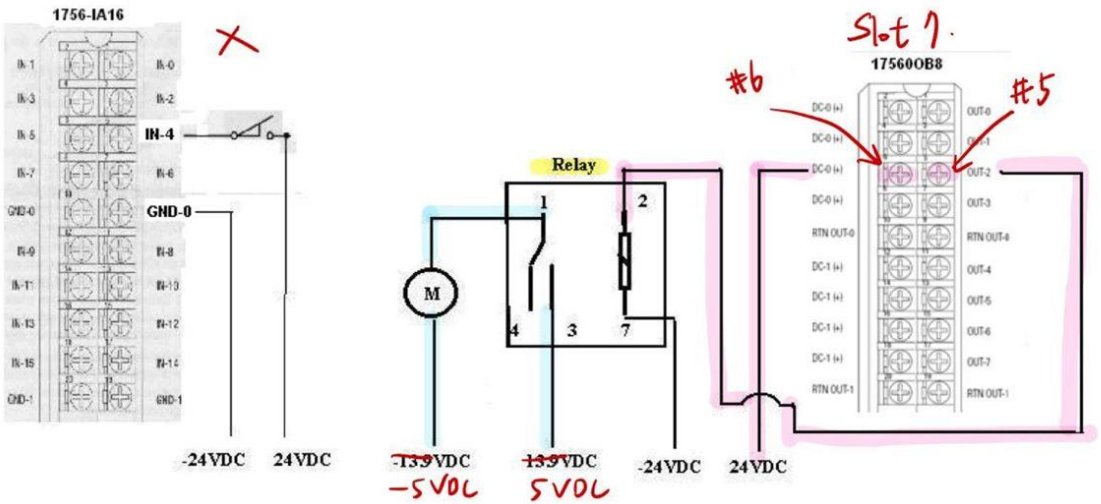
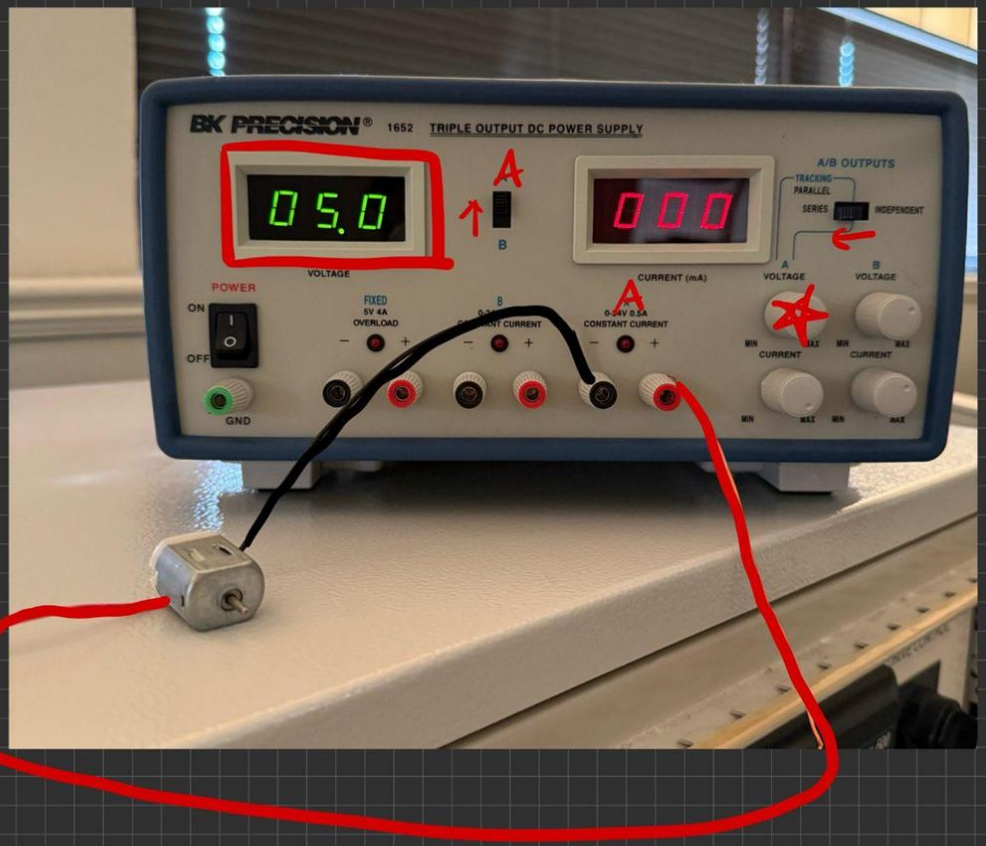
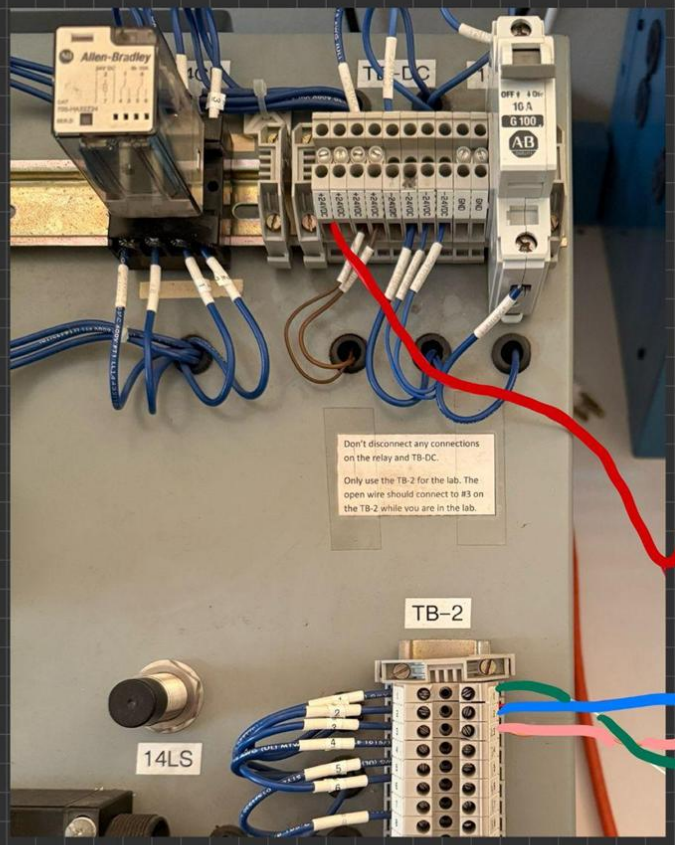


Figure1 shows how the relay, motor, and 1756OB8 module been interfaced.

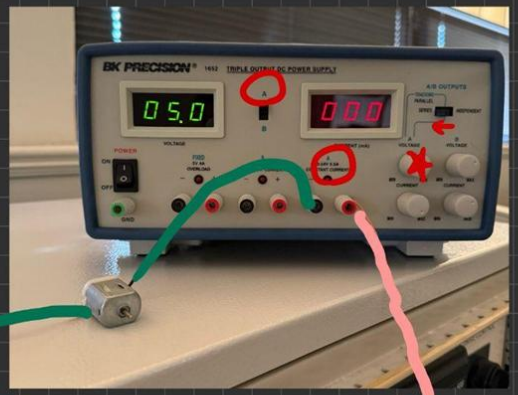
Task 4 a


BK PRECISION 1652 TRIPLE OUTPUT DC POWER SUPPLY
05.0
VOLTAGE
A
B
000
CURRENT (mA)
A/B OUTPUTS
TRACKING
PARALLEL
SERIES
INDEPENDENT
A VOLTAGE
B VOLTAGE
POWER
ON
OFF
GND
FIXED
5V 4A
OVERLOAD
CONSTANT CURRENT
CURRENT
MIN
MAX

Task 4b.


Allen-Bradley
TB-DC
10A
G100
Don't disconnect any connections on the relay and TB-DC.
Only use the TB-2 for the lab. The open wire should connect to #3 on the TB-2 while you are in the lab.
TB-2
14LS


ANALOG OUTPUT
DC OUTPUT
AC INPUT
RELAY OUTPUT


B&K PRECISION
TRIPLE OUTPUT DC POWER SUPPLY
05.0
000

Task 1 - 5

- Task#1 Task#2

a. NO 1-3 6-8 Switch 4 on, light on

NC 1-4 5-8 Switch 4 off, light off

Coil 2,7

b. NO 1-4 5-8 Task#4 Task#5 open means open the circuit, so answers a and b should be opposite.

NC 1-3 6-8 Running Not running

Not running Running

NO: 1-3, 6-8

NC: 1-4, 5-8

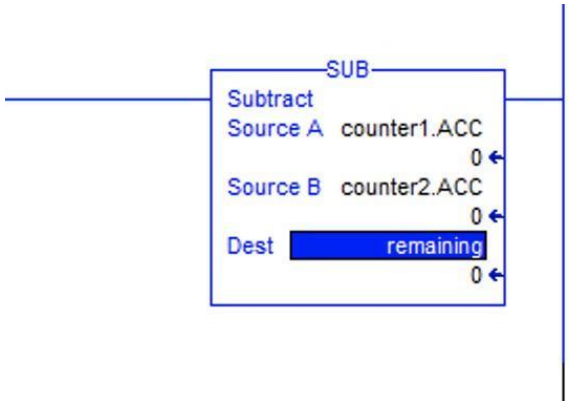
NO: 1-4, 5-8

NC: 1-3,, 6-8

Task 6

Procedures:
click start – system on – conveyor on – (when) IN4 triggered – counter 1 adds 1 – (when) IN5 triggered – counter 2 adds 1 – solenoid on for 2s – click stop – all shut down

How many objects are on conveyor will be stored in SUB (N7:0), the setting for SUB is shown below



Input:

- **start**(IN-0) **stop**(IN-1) **IN4**(IN-4) **IN5**(IN-5)

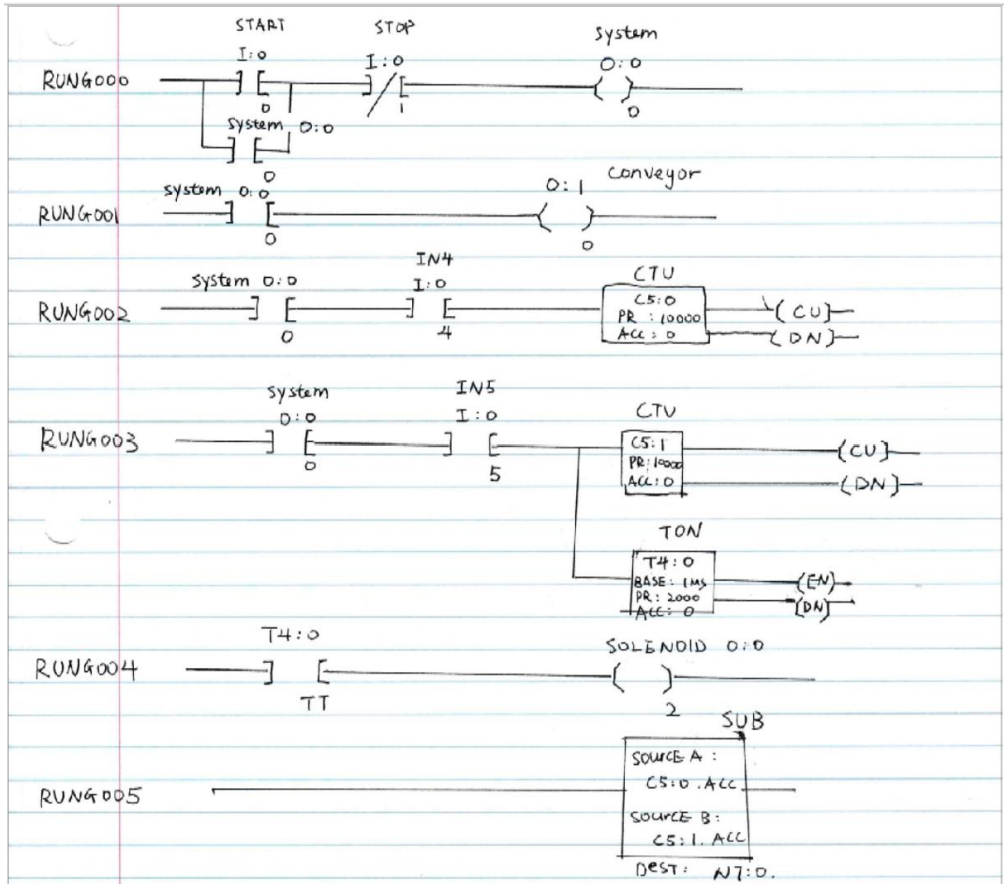
Output:

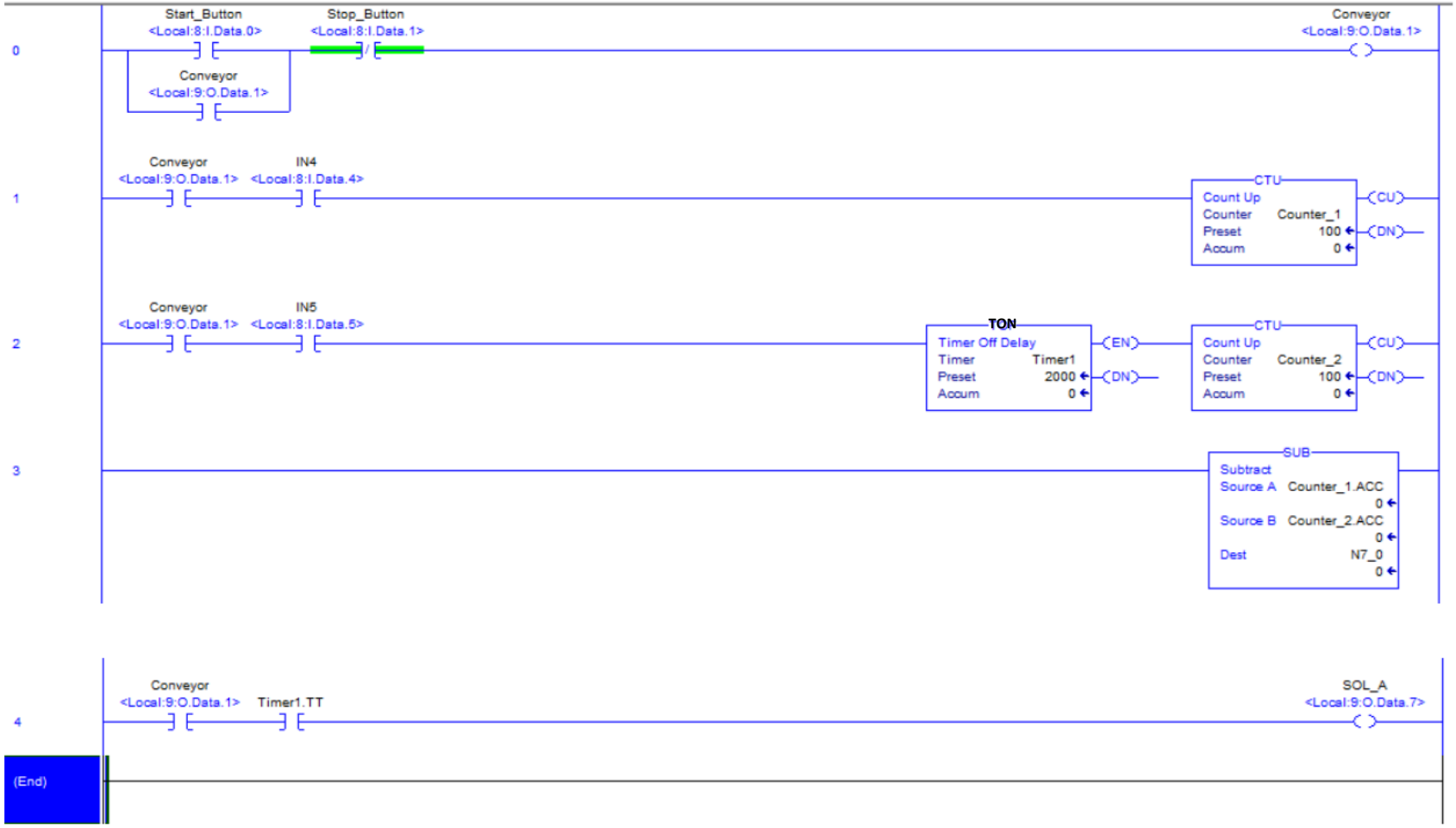
- conveyor(OUT-1) **solenoid**(OUT-2)

Timer/Counter:

- CTU(counter1) CTU(counter2) TON(timer1) SUB(N7:0)
- Note: set large preset values for both CTU

Task 6


RUNG000
START
I:0
0
STOP
I:0
1
System
O:0
0
System O:0
0
RUNG001
system O:0
0
O:1
0
Conveyor
RUNG002
System O:0
0
IN4
I:0
4
CTU
C5:0
PR: 10000
ACC: 0
(CU)
(DN)
RUNG003
system
O:0
0
IN5
I:0
5
CTU
C5:1
PR: 10000
ACC: 0
(CU)
(DN)
TON
T4:0
BASE: 1ms
PR: 2000
ACC: 0
(EN)
(DN)
RUNG004
T4:0
TT
SOLENOID O:0
2
RUNG005
SUB
SOURCE A:
C5:0.ACC
SOURCE B:
C5:1.ACC
DEST: N7:0.


0
Start_Button
<Local:8:I.Data.0>
Stop_Button
<Local:8:I.Data.1>
Conveyor
<Local:9:O.Data.1>
Conveyor
<Local:9:O.Data.1>
1
Conveyor
<Local:9:O.Data.1>
IN4
<Local:8:I.Data.4>
CTU
Count Up
Counter Counter_1
Preset 100
Accum 0
CU
DN
2
Conveyor
<Local:9:O.Data.1>
IN5
<Local:8:I.Data.5>
TON
Timer Off Delay
Timer Timer1
Preset 2000
Accum 0
EN
DN
CTU
Count Up
Counter Counter_2
Preset 100
Accum 0
CU
DN
3
SUB
Subtract
Source A Counter_1.ACC
0
Source B Counter_2.ACC
0
Dest N7_0
0
4
Conveyor
<Local:9:O.Data.1>
Timer1.TT
SOL_A
<Local:9:O.Data.7>
(End)